

The Organizational Development Index (ODI)

Measuring NHL Organizational Player Development, 2010–2020

The Problem

People talk about which NHL teams are good at developing players the same way they talk about anything else in the sport: with a lot of confidence, and not much behind it. A team that identifies a future star and a team that actually develops one into a star get treated as the same thing, even when a trade means most of that player's value shows up somewhere else entirely. Nobody's been checking. That's the part that bothered me enough to build this.

What ODI Does

I built the Organizational Development Index to answer that directly: how efficiently does an NHL organization convert the players it drafts into NHL value relative to what their draft position predicts, and how much of that value does the organization actually keep versus lose through trades. The second half of that question is the real contribution here. I split every player's value two ways, Development Credit, which credits the drafting team no matter what happens afterward, and In-Organization Credit, which only counts what happened while the player was actually still on that roster. That distinction changes real rankings, not just in theory.

Method

I did not simply build this once and trust it. Here's how it was actually built and checked:

- 2,300+ NHL Entry Draft selections, 2010–2020, collected through a Python web scraper I built myself
- Player value measured using Hockey-Reference's Point Shares metric, in fixed 6-year and 8-year post-draft evaluation windows
- An Expected Value curve built directly from this project's own data, adjusted for draft position and player position (forward, defense, goalie)
- Results checked against two independent evaluation windows and a sensitivity test on the model's key assumptions, not just trusted on the first pass

Key Findings

- Anaheim, Boston, and Carolina consistently convert draft picks into more value than expected, across both evaluation windows
- Nashville's ranking collapses once traded players Seth Jones and Samuel Girard's post-trade production is excluded, a 16-point efficiency swing
- Washington's case is sharper: Filip Forsberg was traded before ever playing a game for the organization, so whatever credit they get for him is really just a scouting call, not development
- Detroit's poor ranking, built on the largest sample size in the study (94 picks), is a real signal, not a fluke, unlike Vegas's similarly low rank, which is really just a result of only having 3-4 completed draft classes as a 2017 expansion team

- The positional breakdowns work as a real diagnostic tool: Chicago shows strong forward development (3.46 efficiency) sitting right next to the weakest defensive development in the entire study (-0.54)

Why It Matters

This isn't just a ranking. It's something an organization could actually look at and say, we need to get better here, backed by real numbers instead of a gut feeling or a win total that reflects roster strategy as much as development. It doesn't claim a team that trades a player away gets zero credit for the years it actually had him, only that credit should follow where a player's value was actually produced, not stay permanently with whoever drafted him. It also proves something I believe about hockey as a whole: there's more of this sport that can be measured than people assume, you just have to actually go build it instead of deciding it can't be done.

Learn More

The full paper, methodology, limitations, and source code are available in the links below.